T3 A_+0: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_+0

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y1Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

General-interaction coupling index

Coupling	Field content / definition	Contraction / source

$\lambda^{(1)}_{X_3}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	`del[SU2L @ S2, #3, #4]` `lambdaT3GeneralQ000211I1`
λ_{X_4}	$(S_1^\dagger)^4$	`1` `lambdaT3GeneralQ000400I1`
λ_{X_6}	$S_1\,(S_1^\dagger)^3$	`1` `lambdaT3GeneralQ001300I1`
λ_{X_9}	$(H^\dagger)^2\, S_1\, S_2$	`delS2[SU2L, #4, #1, #2]` `lambdaT3GeneralQ021010I1`
$\lambda_{X_{13}}$	$H\, H^\dagger\, (S_1^\dagger)^2$	`del[SU2L @ fund, #2, #1]` `lambdaT3GeneralQ110200I1`
$\mu_{\mathrm{T3GeneralS1Sq}}$	General scalar bilinear / mass parameter	`muT3GeneralS1Sq`
y_1^{conj}	Additional conjugate-scalar Yukawa	`y1Conj`

Each X_i labels one field-content class in the general-quartic registry; a superscript distinguishes different invariant contractions where present. The original RGBeta identifier and contraction are retained in the index. This is notation only, not a physical normalisation mapping.

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling}(\text{lambdaT3ExtraM000200I1}, \{\}$

$+\text{Coupling}(\lambda_{X_4}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^4$

$+\text{Coupling}(\lambda_{X_{13}}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^2\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{CG}(\text{T3ExtraCGQ110200I1}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))$

$+\text{Bar}(\text{Coupling}(\lambda_{X_6}, \{\}, 0))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQ021010I1}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$

$+\text{Coupling}(\lambda_{X_6}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^3$

$+\text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})))$

$+\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund})))\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))$

$+\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{CG}(\text{T3MixCG}, \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$

$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraM000200I1}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2$

$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^2$

$+\text{Bar}(\text{Coupling}(\lambda_{X_{13}}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{CG}(\text{T3ExtraCGQ110200I1}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))$

$+\text{Bar}(\text{Coupling}(\lambda_{X_6}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^3\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))$

$+\text{Bar}(\text{Coupling}(\lambda_{X_4}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^4$

$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$

$+\text{Bar}(\text{Coupling}(\text{lambdaT3}, \{\}, 0))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{Bar}(\text{T3MixCG}), \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$

$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})))$

$+\text{Coupling}(\lambda_{X_6}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{T3ExtraCGQ021010I1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$

$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$

$+\text{Coupling}(\lambda^{(1)}_{X_3}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^2\text{CG}(\text{T3ExtraCGQ000211I1}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))$

$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$

$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$

$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$

$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$

$+\text{Coupling}(\text{lambda12Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))$

$+\text{Bar}(\text{Coupling}(\lambda^{(1)}_{X_3}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2\text{CG}(\text{T3ExtraCGQ000211I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))$

$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar2SdfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$

$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar2SdfCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$

$-\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$

$+\text{Coupling}(y2, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Y2CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$

$+\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{Coupling}(y_l^{\text{conj}}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{T3Y1ConjCG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(1)), \text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})))$

$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Coupling}(y1, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{T3Y1CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(1)), \text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})))$

$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$

$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Coupling}(y2, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{Bar}(\text{T3Y2CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(-1)), \text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\}))$

$+\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))\text{Bar}(\text{Coupling}(y1, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{Bar}(\text{T3Y1CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2}))\})\text{NCM}(\text{Transp}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$

$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Coupling}(y_l^{\text{conj}}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{Bar}(\text{T3Y1ConjCG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund}))), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2}))\})\text{NCM}(\text{Transp}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L}(\text{fund})), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa1ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`

- HiggsPortal2Inv1
- HiggsPortal2Inv2
- ScalarCrossPortalInv1
- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_2_1_1_1
- T3ExtraQ_0_0_0_4_0_0_1
- T3ExtraQ_0_0_1_3_0_0_1
- T3ExtraQ_0_2_1_0_1_0_1
- T3ExtraQ_1_1_0_2_0_0_1
- T3ExtraM_0_0_0_2_0_0_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{59{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$
\begin{aligned}
16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\
& + \frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 2\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\
& - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}
\end{aligned}
$$

y_1

$$
\begin{aligned}
16\pi^2 \frac{dy_1}{d\ln\mu} = & 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\Big)\,(y_1)_{ij} \\
& + 4\,\mathrm{Tr}\Big(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\Big)\left(y_1^{\mathrm{conj}}\right)_{ij} \\
& + 2T_\nu^{(1)}\,(y_1)_{ij} \\
& + \frac{1}{4}(10\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_1^{\mathrm{conj}}\right)_{ij} + 10\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_1\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 4\left(y_1\ (y_1)^\dagger\ y\right. \\
& - \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_1)_{ij}
\end{aligned}
$$

y_2

$$
\begin{aligned}
16\pi^2 \frac{dy_2}{d\ln\mu} = & T_\nu^{(2)}\,(y_2)_{ij} \\
& + \frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_2)_{ij}
\end{aligned}
$$

M_F

$$
\begin{aligned}
16\pi^2 \frac{dM_F}{d\ln\mu} = & \frac{1}{4}(2(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(\left(y_1^{\mathrm{conj}}\right)^T\ \left(y_1^{\mathrm{conj}}\right)^*\ M_F\right)_{ij}) + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(M_F)_{ij}
\end{aligned}
$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & m_1^2(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 4\lambda_{S_1}^{(1)}) \\ & + \mu_{\mathrm{T3GeneralS1Sq}}(8\,\mathrm{Tr}\Big(y_1\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 12\,(\lambda_{X_6})^*) \\ & - 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big) \\ & - 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ \Big(y_1^{\mathrm{conj}}\Big)^T\ \Big(y_1^{\mathrm{conj}}\Big)^*\Big) \\ & + 8\,(\mu_{\mathrm{T3GeneralS1Sq}})^*\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big) \\ & + 12\lambda_{X_6}\,(\mu_{\mathrm{T3GeneralS1Sq}})^* \\ & + 6\lambda_{12}^{(1)} m_2^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)}) \\ & - 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big) \\ & + 4\mu_{\mathrm{T3GeneralS1Sq}}\left(\lambda_{X_3}^{(1)}\right)^* \\ & + 4\lambda_{X_3}^{(1)}\,(\mu_{\mathrm{T3GeneralS1Sq}})^* \\ & - 12{g_2}^2 m_2^2 \\ & - 6{g_1}^2 m_2^2 \\ & + 2\lambda_{12}^{(1)} m_1^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\,(\lambda_5)^* \\ & + 4\lambda_{X_9}\,(\lambda_{X_9})^* \\ & + 8\lambda_{X_{13}}\,(\lambda_{X_{13}})^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1 \\ & + \frac{3{g_1}^4}{4} \\ & + 12{\lambda_1}^2 \\ & + 2{\lambda_{HS_1}^{(1)}}^2 \\ & + 6{\lambda_{HS_2}^{(1)}}^2\end{aligned}$$

$$+ 3{\lambda^{(A)}_{HS_2}}^2$$

$\lambda^{(1)}_{S_1}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda^{(1)}_{S_1}}{d\ln\mu} = \lambda^{(1)}_{S_1}(8T^{(1)}_{\nu} &+ 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right))\\
&+ \lambda_{X_6}(24\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right) + 108\,(\lambda_{X_6})^*)\\
&+ 24\,(\lambda_{X_6})^*\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^{\dagger}\right)\\
&- 16\,\mathrm{Tr}\left(y_1\ (y_1)^{\dagger}\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)\\
&- 16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\ y_1^{\mathrm{conj}}\ (y_1)^{\dagger}\right)\\
&- 8\,\mathrm{Tr}\left(y_1\ (y_1)^{\dagger}\ y_1\ (y_1)^{\dagger}\right)\\
&- 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^{\dagger}\right)\\
&+ 12\lambda^{(1)}_{X_3}\left(\lambda^{(1)}_{X_3}\right)^*\\
&+ 288\lambda_{X_4}\,(\lambda_{X_4})^*\\
&+ 8\lambda_{X_{13}}\,(\lambda_{X_{13}})^*\\
&+ 6{\lambda^{(1)}_{12}}^2\\
&+ 4{\lambda^{(1)}_{HS_1}}^2\\
&+ 10{\lambda^{(1)}_{S_1}}^2
\end{aligned}$$

$\lambda^{(1)}_{S_2}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda^{(1)}_{S_2}}{d\ln\mu} = \lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} &+ 8\lambda^{(A)}_{S_2})\\
&- \frac{8}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^{\dagger}\ y_2\ (y_2)^{\dagger}\right)\\
&+ 8\lambda^{(1)}_{X_3}\left(\lambda^{(1)}_{X_3}\right)^*\\
&+ 22{g_2}^4\\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda^{(1)}_{S_2}\right)\\
&- 24{g_2}^2\lambda^{(1)}_{S_2}\\
&+ 12{g_1}^4\\
&+ 2{\lambda^{(1)}_{12}}^2\\
&+ 4{\lambda^{(1)}_{HS_2}}^2\\
&+ 14{\lambda^{(1)}_{S_2}}^2\\
&+ 16{\lambda^{(A)}_{S_2}}^2
\end{aligned}$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + \lambda_{X_{13}}(8\,\mathrm{Tr}\Big(y_1\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 4(3\,(\lambda_{X_6})^* + 4\,(\lambda_{X_{13}})^*)) \\
& + 8\,(\lambda_{X_{13}})^*\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big) \\
& - 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 12\lambda_5\,(\lambda_5)^* \\
& + 12\lambda_{X_6}\,(\lambda_{X_{13}})^* \\
& + 12\lambda_{X_9}\,(\lambda_{X_9})^* \\
& - \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
& - \frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& - 2\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 4\lambda_{X_{13}}\,\Big(\lambda_{X_3}^{(1)}\Big)^* \\
& + 4\lambda_{X_3}^{(1)}\,(\lambda_{X_{13}})^* \\
& + 4\lambda_{X_9}\,(\lambda_{X_9})^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 3{g_1}^4 \\
& - \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
& + 6\lambda_1\lambda_{HS_2}^{(1)} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & \lambda_{12}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)}) \\
& + \lambda_{X_3}^{(1)}(8\,\mathrm{Tr}\Big(y_1\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 4(4\left(\lambda_{X_3}^{(1)}\right)^* + 3\,(\lambda_{X_6})^*)) \\
& + 8\left(\lambda_{X_3}^{(1)}\right)^* \mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big) \\
& - 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 12\lambda_{X_6}\left(\lambda_{X_3}^{(1)}\right)^* \\
& + 4\lambda_{X_9}\,(\lambda_{X_9})^* \\
& - 12{g_2}^2\lambda_{12}^{(1)} \\
& - 6{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = {} & \lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)}) \\
& + 4\,(\lambda_{X_9})^*\,\mathrm{Tr}\Big(y_1\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) \\
& + 4\,(\lambda_{X_9})^*\,(\left(\lambda_{X_3}^{(1)}\right)^* + 2\,(\lambda_{X_{13}})^*) \\
& - \frac{21{g_2}^2\lambda_5}{2} \\
& - \frac{9{g_1}^2\lambda_5}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_5 \\
& + 2\lambda_1\lambda_5 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_5 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\
& + \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + \frac{8}{3}\lambda_5\,(\lambda_5)^* \\
& + \frac{8}{3}\lambda_{X_9}\,(\lambda_{X_9})^*
\end{aligned}
$$

$$
\begin{aligned}
&+ {g_1}^2 \left(4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2} \right) \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &\, 4\lambda_{S_2}^{(A)} T_\nu^{(2)} \\
&- \frac{2}{3}\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger \Big) \\
&- 2{g_2}^4 \\
&+ {g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_2}^{(A)} \right) \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

y_1^{conj}

$$
\begin{aligned}
16\pi^2 \frac{dy_1^{\text{conj}}}{d\ln\mu} = &\, 4\operatorname{Tr}\Big(y_1^{\text{conj}}\ (y_1)^\dagger \Big) (y_1)_{ij} \\
&+ 2T_\nu^{(1)} \left(y_1^{\text{conj}} \right)_{ij} \\
&+ 2\operatorname{Tr}\Big(y_1^{\text{conj}}\ \left(y_1^{\text{conj}} \right)^\dagger \Big) \left(y_1^{\text{conj}} \right)_{ij} \\
&+ \frac{1}{4}(10 \left(y_1\ (y_1)^\dagger\ y_1^{\text{conj}} \right)_{ij} + 10 \left(y_1^{\text{conj}}\ (y_1)^\dagger\ y_1 \right)_{ij} + 3 \left(y_2\ (y_2)^\dagger\ y_1^{\text{conj}} \right)_{ij} + 2 \left(Y_e\ (Y_e)^\dagger\ y_1^{\text{conj}} \right)_{ij} + 4 \Big(y_1^{\text{conj}}\ \Big(y_1^{\text{conj}} \\
&- \frac{9}{2}{g_2}^2 \left(y_1^{\text{conj}} \right)_{ij} \\
&- \frac{3}{2}{g_1}^2 \left(y_1^{\text{conj}} \right)_{ij}
\end{aligned}
$$

$\lambda_{X_3}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_3}^{(1)}}{d\ln\mu} = &\, \lambda_{12}^{(1)}(4\operatorname{Tr}\Big(y_1^{\text{conj}}\ (y_1)^\dagger \Big) + 8\lambda_{X_3}^{(1)} + 6\lambda_{X_6}) \\
&+ \lambda_{X_3}^{(1)}(4T_\nu^{(1)} + 4\operatorname{Tr}\Big(y_1^{\text{conj}}\ \left(y_1^{\text{conj}} \right)^\dagger \Big) + 2T_\nu^{(2)}) \\
&- 4\operatorname{Tr}\Big(y_1^{\text{conj}}\ (y_1)^\dagger\ y_2\ (y_2)^\dagger \Big) \\
&+ 4\left(\lambda_5\right)^* \left(\lambda_{X_9}\right)^*
\end{aligned}
$$

$$
\begin{aligned}
&+ 24\lambda_{X_4} \left(\lambda_{X_3}^{(1)}\right)^* \\
&- 12{g_2}^2\lambda_{X_3}^{(1)} \\
&- 6{g_1}^2\lambda_{X_3}^{(1)} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{X_{13}} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{X_3}^{(1)} \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{X_3}^{(1)} \\
&+ 4\lambda_{S_2}^{(A)}\lambda_{X_3}^{(1)}
\end{aligned}
$$

λ_{X_4}

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_4}}{d\ln\mu} = &\lambda_{X_4}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)) \\
&+ 4\lambda_{X_6}\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) \\
&- 4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{X_4} \\
&+ 3{\lambda_{X_3}^{(1)}}^2 \\
&+ 9{\lambda_{X_6}}^2 \\
&+ 2{\lambda_{X_{13}}}^2
\end{aligned}
$$

λ_{X_6}

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_6}}{d\ln\mu} = &\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) + 18\lambda_{X_6}) \\
&+ \lambda_{X_4}(16\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + 72\,(\lambda_{X_6})^*) \\
&+ \lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right)) \\
&- 8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ (y_1)^\dagger\right) \\
&- 8\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) \\
&+ 6\lambda_{12}^{(1)}\lambda_{X_3}^{(1)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{X_{13}}
\end{aligned}
$$

λ_{X_9}

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_9}}{d\ln\mu} = &\,4\,(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) \\
&+ \lambda_{X_9}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(y_1^{\mathrm{conj}}\ \left(y_1^{\mathrm{conj}}\right)^\dagger\right) + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)) \\
&+ 4\,(\lambda_5)^*\,(\left(\lambda_{X_3}^{(1)}\right)^* + 2\,(\lambda_{X_{13}})^*) \\
&- \frac{21{g_2}^2\lambda_{X_9}}{2} \\
&- \frac{9{g_1}^2\lambda_{X_9}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{12}^{(1)}\lambda_{X_9} \\
&+ 2\lambda_{1}\lambda_{X_9} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{X_9} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{X_9} \\
&+ 4\lambda_{HS_2}^{(A)}\lambda_{X_9}
\end{aligned}
$$

$\lambda_{X_{13}}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_{13}}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big) + 6\lambda_{X_6} + 8\lambda_{X_{13}}) \\
& + \lambda_{X_{13}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 12\,(\lambda_5)^*\,(\lambda_{X_9})^* \\
& + 24\lambda_{X_4}\,(\lambda_{X_{13}})^* \\
& - \frac{9{g_2}^2\lambda_{X_{13}}}{2} \\
& - \frac{3{g_1}^2\lambda_{X_{13}}}{2} \\
& + 6\lambda_1\lambda_{X_{13}} \\
& + 6\lambda_{HS_2}^{(1)}\lambda_{X_3}^{(1)} \\
& + 2\lambda_{S_1}^{(1)}\lambda_{X_{13}}
\end{aligned}
$$

$\mu_{\mathrm{T3GeneralS1Sq}}$

$$
\begin{aligned}
16\pi^2 \frac{d\mu_{\mathrm{T3GeneralS1Sq}}}{d\ln\mu} = {} & m_1^2(4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ (y_1)^\dagger\Big) + 6\lambda_{X_6}) \\
& - 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ \Big(y_1^{\mathrm{conj}}\Big)^T\ (y_1)^*\Big) \\
& + \mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(y_1^{\mathrm{conj}}\ \Big(y_1^{\mathrm{conj}}\Big)^\dagger\Big)) \\
& + 24\lambda_{X_4}\,(\mu_{\mathrm{T3GeneralS1Sq}})^* \\
& + 2\lambda_{S_1}^{(1)}\mu_{\mathrm{T3GeneralS1Sq}} \\
& + 6\lambda_{X_3}^{(1)}m_2^2
\end{aligned}
$$